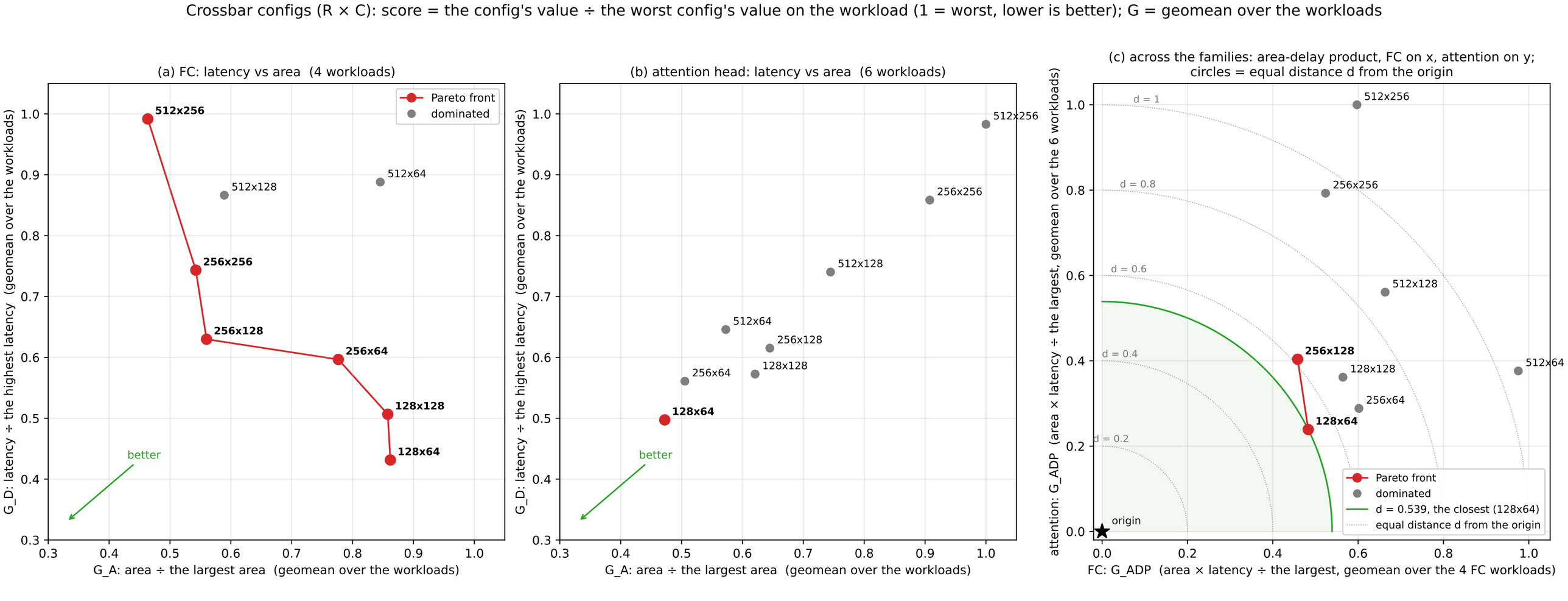
Crossbar configs (R × C): score = the config's value ÷ the worst config's value on the workload (1 = worst, lower is better); G = geomean over the workloads
(a) FC: latency vs area (4 workloads)
Pareto front
dominated
512x256
512x128
512x64
256x256
256x128
256x64
128x128
128x64
better
G_D: latency ÷ the highest latency (geomean over the workloads)
G_A: area ÷ the largest area (geomean over the workloads)
(b) attention head: latency vs area (6 workloads)
512x256
256x256
512x128
512x64
256x128
128x128
256x64
128x64
better
G_D: latency ÷ the highest latency (geomean over the workloads)
G_A: area ÷ the largest area (geomean over the workloads)
(c) across the families: area-delay product, FC on x, attention on y;
circles = equal distance d from the origin
d = 1
d = 0.8
d = 0.6
d = 0.4
d = 0.2
512x256
256x256
512x128
512x64
256x128
128x128
256x64
128x64
origin
Pareto front
dominated
d = 0.539, the closest (128x64)
equal distance d from the origin
attention: G_ADP (area × latency ÷ the largest, geomean over the 6 workloads)
FC: G_ADP (area × latency ÷ the largest, geomean over the 4 FC workloads)